Parallel Morphological Image Processing

Accelerating morphological operations via C++17 and OpenMP

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Introduction

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- **Mathematical Morphology:** framework for geometric feature extraction and image restoration
- **Targeted Operations:**
  - *Fundamental*: Erosion, Dilation
  - *Compound*: Opening, Closing
- **Goal:** resolve computational bottlenecks in high-resolution image processing
- **Technology Stack:** `C++17` and `OpenMP` for thread orchestration
- **Dataset & Benchmarks:** BSDS500
  - Preliminary grayscale conversion
  - 2.0x and 4.0x upscaling to evaluate scalability

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Context & Architectural Limits

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Operator Logic

- **Erosion:** minimum extraction in a defined spatial neighborhood
- **Dilation:** maximum extraction in a defined spatial neighborhood
- **Opening:** Erosion followed by Dilation
- **Closing:** Dilation followed by Erosion

The Memory Wall & Complexity

- Naive execution with 2D stencils strictly bound by $\mathcal{O}(M \times N \times K^2)$ complexity
- **Compound operations** implemented sequentially (double-pass approach)
- **System Overhead:** destruction of *temporal locality* and cache saturation between passes

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Hardware Optimization: Pixel-Level Parallelism

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- Spatial stencil processing is inherently independent for each pixel
- **Loop Collapsing (`collapse(2)`):** logical fusion of X and Y image axes
  - Expands the iteration space and distributes workload with fine granularity
  - Prevents *thread starvation* at boundaries
- **Memory Safety (`default(none)`):** forced explicit scoping
  - Thread-local allocation for temporary variables (e.g., `min_val`)
  - Avoids synchronization overhead (zero locks or atomics) in the critical loop
- **Static Scheduling (`schedule(static)`):** deterministic work distribution
  - Zero runtime overhead, ideal for uniform per-pixel workloads (invariant $K \times K$ footprint)
  - *Note*: dynamic chunking later explored

Algorithmic Optimization: 1D Separable Kernels

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- **The Dominant Lever:** decomposition of the $K \times K$ stencil into two independent passes (Horizontal $\rightarrow$ Vertical)
- **Computational Reduction:** transition from $\mathcal{O}(K^2)$ to linear $\mathcal{O}(2K)$
  - Using a 15×15 kernel, base operations per pixel drop from 225 to just 30

The Architectural Trade-off (Cache Asymmetry)

- **Horizontal Pass:** highly efficient, exploits prefetching and RAM *spatial locality* (row-major layout)
- **Vertical Pass:** introduces severe cache misses, forcing the memory bus to work column-wise

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Producer-Consumer Pipeline

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- **The Fork-Join Problem:** pure 1D Closing requires 4 global passes, doubling heap footprint and introducing strict synchronization barriers
- **OpenMP Pipeline Design:**
  - Single partitioned parallel region where half the threads act as Producers and half as Consumers
  - Fine-grained chunk synchronization via `PaddedFlag` array to eliminate false sharing
- **Architectural Benefits:** execution overlap and keeps pixels in cache (*temporal locality*)

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Benchmarking Methodology

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- **Target Hardware:** Rigorous execution on a physical Quad-Core CPU to limit fictitious hyper-threading influence
- **Dataset:** 200 heterogeneous images extracted from BSDS500
- **Automated Pre-processing:** Python pipeline for uniform conversion and upscaling
- **Evaluation Metrics:**
  - *Strong Scaling:* Maintaining workload (2x/4x) while scaling physical thread count
  - *Weak Scaling:* Balanced and proportional increase of computing resources and resolution (data volume)

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Performance Results

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Strong & Weak Scalability Analysis

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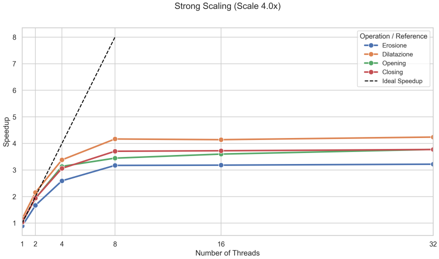

Figure: Strong Scaling (Fixed Workload)

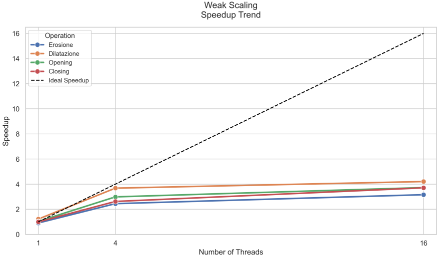

Figure: Weak Scaling (Scaled Workload)

- Asymptotic saturation at **3.5x–4.0x** with 4 physical cores
- Perfectly reflects *Amdahl's Law* (unavoidable pure sequential fraction)

- Constant execution times when doubling workload and cores
- Confirms excellent hardware resource distribution

Algorithmic Reduction: 1D vs 2D Performance
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- **Direct Impact:** transitioning to linear $\mathcal{O}(2K)$ complexity acts as the true performance multiplier
- **Kernel Scaling (K):**
  - At low K values, the gap is minimal
  - At high K values, the 2D approach collapses under quadratic load

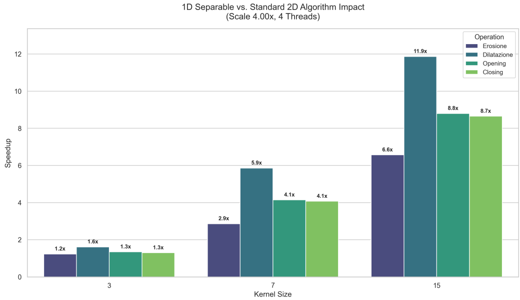

Figure: Speedup of 1D Separable Kernels vs 2D baseline

Producer-Consumer Pipeline Efficiency

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- **Focus:** specific optimization for Opening and Closing
- **Peak Efficiency:** yields up to a **1.67x** performance boost over the standard parallel baseline
- **High-Concurrency Contention:**
  - Active *spin-wait* loops saturate physical cores under high logical thread counts
  - Cross-chunk dependencies amplify synchronization overhead and cause thread starvation

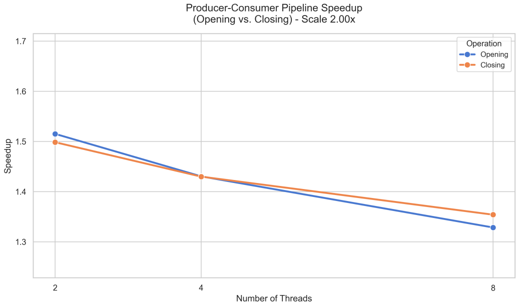

Figure: Pipeline speedup for compound operations

Other Strategies Tested

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- **SIMD Vectorization (`#pragma omp simd`):**
  - No positive impact on core loops
  - Failure due to compiler auto-vectorization (already optimal) and min/max conditional branches
- **Dynamic Loop Scheduling (`schedule(dynamic)`):**
  - Evaluated on the baseline parallel version to improve load balancing
  - Performance remained virtually identical to the default `static` approach

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Peak Performance & Future Developments

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Optimal Configuration

22x Peak Speedup

Mixed 1D Separable + OpenMP architecture ($K = 15$, 4 Threads) over 2D Baseline.

- **Hardware/Software Conclusion:** Bare multithreading cannot save a flawed algorithm. Hardware/software integration is the only path to High-Performance
- **Heterogeneous Computing (Future Work):**
  - Porting the logic to GPU architectures (CUDA)
  - The SIMT model and hardware Shared Memory would resolve vertical memory cache asymmetries

Visual Results (BSDS500)

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Original

Erosion

Dilation

Opening

Closing

Parallel Morphological Image Processing

Thank you for listening!

Any questions?